Approval-Based Elections in the 1D Euclidean Domain

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This dissertation is submitted for the degree of
Master of Science

April 2021

TODO

0.1 Resources

https://github.com/jasieksz/master-thesis-code/blob/master/flat-code/ch8_main.py

0.2 My Questions

1. Fitzsimmons Lackner 2019, COP - SAT Solver

0.3 10/12/2020

1. Refactor code - vcr ilp detection, profile generation
2. Generate and analyze profiles: 5x5, 4x6, 6x4, random NxM

0.4 17/12/2020

1. Implement ilp MAV
2. Implement VR/CR MAV **?**
3. Compare ilp and poly MAV

0.5 07/01/2021

1. VR/CR MAV **?**

0.6 14/01/2021

1. VR/CR MAV **?**

0.7 11/02/2021

1. write thesis (+- 3 pages)
2. email about CR MAV to Liu/Guo
3. fix CR MAV algo
4. try to proof CR MAV
5. single thread experiments

0.8 18/02/2021

1. investigate CR MAV
2. VCR MAV ideas

3. email about CR MAV to Liu/Guo
4. try to proof CR MAV
5. write thesis
6. 20x20 VCR profiles - random distribution analysis
7. delete some random Candidates for VCR profile to become VR / CR

0.9 25/02/2021

1. investigate CR MAV
2. VCR MAV ideas
3. write thesis

0.10 04/03/2021

1. Chińczycy - wyslac kontrprzykład
2. VCR MAV ideas
3. write thesis - ch.3 (1D Euclidean Pref.) and ch.5 (domain analysis)
4. 20x20 VCR profiles - random distribution analysis
5. delete some random Candidates for VCR profile to become VR / CR
6. Control problem for VCR
7. Justified Representation (Proportional / Extended) for VCR

0.11 11/03/2021

1. VCR MAV ideas
2. write thesis - ch.6 MAV

0.12 18/03/2021

1. research C1P - NP proofs
2. PoC for deleting C's and/or V's for VCR -> CR / VR
3. research Control Problem
4. fix ch.4 and 5, send for the review (monday)

0.13 08/04/2021

1. bigO: *av-cc-dv* naive greedy
2. ch4 (detection) full draft

Abstract

(WIP)

I study approval-based elections, where preferences are in the 1D-Euclidean domain. I explore multiple areas, notably:

1. I create domain detection algorithms and evaluate the results.
2. I look for the efficient method of computing election results.

For the first issue I find ???. For the second one I create polynomial complexity algorithm (not yet) and compare it with LP solutions. (WIP)

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Chapter 1

Introduction

Elections are ubiquitous element of our society in one form or another. For example choosing members of parliament or a winner of a competition. In a day-to-day activities we need to decide on a meeting time or select recommended movies in a streaming service. Each of these choices may require different type of election and scoring rule, based on the underlying properties of the choice (i.e. voters should be represented proportionally by members of a government)

Reseach objectives

In this work I am going to focus on the *Approval Elections*, that is where each voter provides a subset of candidates that he approves of and disapproves the rest. In general many social choice problems are known to be computationally hard. However they become tractable under some restrictions on the structure of the preferences. Elkind and Lackner (2015) explored well known restrictions in the setting of approval preferences, introduced several new one-dimensional domain restrictions such as Dichotomous Euclidean (sprawdzic kiedy/kto wprowadził pojęcie - spatial pref, multidim unfolding) and derivatives (czy to dobre okreslenie na VR CR?). Faliszewski (2021) generalized this model for the N-dimensional euclidean space calling it Voter Candidate Range and explored two-dimensional case.

The goal of this work is to explore properties of the one-dimensional model. Search for the profile detection polynomial complexity algorithm. I will try as well to extend solution of **?** for computing MAV score with polynomial complexity to the VCR model.

Contributions

I provide linear programming preference detection algorithm and ...

Chapter 2

Preliminaries

In this chapter, we introduce the definitions and notations used in this work.

Approval Elections

Election is a pair $E = (C, V)$, where $C = \{c_1, ..., c_m\}$ is a set of candidates and $V = \{v_1, ..., v_n\}$ is a set of voters. Each voter $v \in V$ approves a subset of the candidates and disapproves the remaining candidates. I will be referring to it as a *approval set* and for each voter $v \in V$ denote it as $A(v)$, when using A just by itself I refer to all approval sets in the given election (preferences).

Committee Selection Rules

Voting rules are algorithms used to select winners (committee) of a given election. Each rule has different properties and must be selected accordingly to the goal of the election. For the detailed overview of multiwinner voting rules I point to Faliszewski et al. (2017).

WIP: Jakie jeszcze tematy/pojęcia tu opisać

Integer Linear Program

Integer linear programming (ILP) is a common approach to solve NP-hard problems, furthermore it can be used effectively for other problems as well. (WIP: czy zdanie od furthermore ... jest potrzebne) To define a computational problem as an integer linear program, we have to create a system of linear inequalities (constraints) (WIP: Sprawdzic czy na pewno inequalities == constraints) and optionally a linear function to optimize, alternatively we can just check feasibility of the constraints. Next to obtain a solution we have to use an optimization solver. Popular choices are Gurobi or CPLEX which are state-of-the-art general purpose solvers.

(WIP: Czy przenieść do Preliminaries + opisać rodzaje zmiennych w ILP, etc. ?)

Chapter 3

One Dimmensional Euclidean Domain

3.1 Preference Restrictions

(WIP: Motywacja, po co restrykcje, czy moge powtorzyc Introduction?)

Many social choice problems such as winner determination, (WIP: Control?) are known to be computationally hard. To deal with it we can use approximate or parametrized algorithms (WIP: Podać przykłady metod / cytacje). The alternative approach is to introduce restrictions on the preference domains. (WIP: Cytacje research'u) In the last few years this area of research advanced a lot, providing multiple efficient algorithms for the popular committee selection rules, and other problems.

3.2 Euclidean Preference Restrictions

Voter Candidate Range

Given an election $E = (C, V)$, we say that the voters have *1D-Euclidean* preferences if for every agent (voter or candidate) there exists a point x_a and nonnegative real value radius r_a such that: $\forall\, c \in C\ \forall\, v \in V : c \in A(v) \Leftrightarrow \sqrt{(x_c - x_v)^2} \leq r_c + r_v$

We can easily think of it as: the point x_a represents ideal position in the 1-dimensional space of opinions. The radius can be seen as candidate's charisma or voter's willingness to compromise.

Candidate Range

We say that P satisfies CR if candidates in C can be ordered so that each of the sets v_i forms an interval of that ordering. CR is special case of VCR, where we require all the voters to have radii equal to zero

Voter Range

We say that P satisfies VR if the voters in V can be reordered so that for every candidate c_i the voters that approve c form an interval of that ordering. VR is a special case of VCR, where we require all the candidates to have radii equal to zero

Voter Interval and Candidate Interval

Faliszewski (2021) showed that CR and VR are equivalent to respectively VI and CI introduced by Elkind and Lackner (2015).

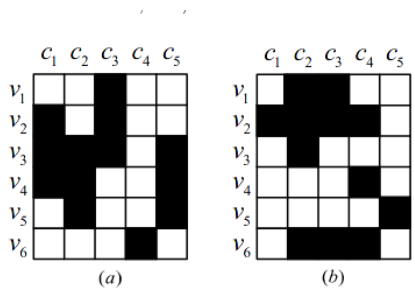

Figure 3.1 CR and VR Temporary image

Larger Domain

(WIP: Zmienić na mój przykład i opisać dlaczego nie vr i nie cr)

The VCR model has larger domain than the VR and CR combined. For example, election with candidate set $C = \{a,b,c,d\}$ and voters with approval sets $\{a,b\}$, $\{b,c\}$, $\{b,d\}$, and $\{a,b,c,d\}$ is VCR, but neither VR nor CR.

Chapter 4

Voter Candidate Range Profile Detection

In this chapter we present our algorithms for deciding if a given approval profile has the VCR property.

4.1 Profile Structure Detection

To take advantage of the restrictions defined in Section 3, we need to design detection algorithms. They can be used to check whether given election profiles satisfy particular structural properties. For example, to check the VR (CR) property it is enough to check for the *consecutive ones property* in the preferences matrix.

Definition 1. *A binary matrix (i.e., a matrix consisting of zeros and ones) has the consecutive ones property (C1P) if its columns be permuted in such a way that for each row the ones are consecutive, i.e., the 1s form an interval.*

In this work we refer to the term *C1P* more broadly, using it interchangeably, for an analogous scenario where we permute the rows and look for the consecutive 1s in every column in a given matrix. Booth and Lueker (1976) provided linear-time C1P detection algorithm based on the PQ-Tree data structure.

Proposition 1. *Elkind and Lackner (2015) Deciding whether an approval profile satisfies VR or CR is possible in $O(m \cdot n)$ time.*

Proof. Let $C = \{c_1, ..., c_m\}$ and $V = \{v_1, ..., v_n\}$, we create a $n \times m$ matrix A (a_{ij}): for each $i \in [n]$ and $j \in [m]$: $a_{ij} = 1$ if $c_j \in ApprovalSet(v_i)$ and 0 otherwise. We can see that for the VR property, permuting the columns of matrix A so that ones form an interval in each row is equivalent to permuting candidates so that the set of candidates approved by each voter forms an interval. For the CR case, we just need to transpose the matrix created for the VR. □

Our focus lies on the one dimensional VCR property. Having preference matrix as input we have to provide for each agent (voter/candidate) a point on the real line and a nonnegative real value representing radius, so that these parameters satisfy the VCR property.

4.2 Integer Linear Program

An algorithm based on the ILP solving is an excellent choice for the first attempt at designing a detection algorithm. It provides an efficient and relatively simple solution. Additionally, it can be used as a benchmark for validating future solutions. Below we provide an ILP for the VCR profile recognition, together with additional preliminary steps.

Transforming the Definition of VCR into Linear Constraints

Before formulating an ILP for detecting VCR profiles, we have to redefine the VCR definition in order to make it appropriate for an ILP solver. One dimensional VCR definition is based on the euclidean distance which cannot be directly represented in the ILP. Equation 4.2 shows transformation of the VCR definition into linear inequalities. (WIP: Czy powtarzam *VCR definition* za często?)

As a short reminder from Section 3.2, for the candidates C and voters V, 1D-VCR is defined as:

$$\forall\, c \in C\, \forall\, v \in V : c \in A(v) \Leftrightarrow \|x_c - x_v\| \leq r_c + r_v \tag{4.1}$$

(WIP: W jaki sposób mogę zwięźle, bez ponownego wpropadzania wszystkich oznaczeń, przypomieć definicję? O ile w ogóle jest to potrzebne...)

$$\begin{aligned} &\|x_c - x_v\| \leq r_c + r_v \\ &\iff -(r_c + r_v) \leq x_c - x_v \leq r_c + r_v \\ &\iff x_c - x_v \leq r_c + r_v \;\; \vee \;\; -(x_c - x_v) \leq r_c + r_v \end{aligned} \tag{4.2}$$

(WIP: Poprawić formatowanie)

The constraint obtained from the equation 4.2 is not suitable yet for the ILP because it contains logical *OR* condition which again cannot be directly represented in an ILP. However, this issue is quite common and most solvers provide additional built-in mechanisms to solve it.

In Gurobi solver it is called *indicator constraint*, which allows creating versatile constraints in form of: $z = f \rightarrow a^T x \leq b$ which states that if the binary indicator variable z is

equal to f, where $f \in \{0,1\}$, then the linear constraint $a^T x \leq b$ should hold. On the other hand, if $z = 1 - f$, the linear constraint may be violated.

Detection algorithm

Final formulation of the ILP for recognising VCR property in the election profile is shown in the figure 4.3. Variables used by the program x_a and r_a represent agent's point on the real line and their radius respectively. Additionally, we use two binary variables p and q required by the indicator constraints.

(WIP: Czy pisać jeszcze raz że approval set A jest jedyną daną wejściową?)

(WIP: Wyjaśnić rodzaje zmiennych w ILP? (binary, etc.))

(WIP: Poprawić format zapisu ILP na bardziej standardowy)

$$
\begin{aligned}
&\textbf{check feasibility}: \\
&\quad \textbf{subject to}: \\
&\qquad x_a \in \mathbb{R} && \forall \text{ a} \in C \cup V \\
&\qquad r_a \in \mathbb{R}^{0+} && \forall \text{ a} \in C \cup V \\
&\qquad p_{cv} \in \{0,1\} && \forall \text{ c} \in \text{C} \; \forall \text{ v} \in V \\
&\qquad q_{cv} \in \{0,1\} && \forall \text{ c} \in \text{C} \; \forall \text{ v} \in V \\
&\quad \text{if } c \in A(v): \\
&\qquad x_c - x_v \leq r_c + r_v \\
&\qquad -(x_c - x_v) \leq r_c + r_v \\
&\quad \text{if } c \notin A(v): \\
&\qquad p_{cv} == 1 \rightarrow x_c - x_v \geq r_c + r_v + 1 \\
&\qquad q_{cv} == 1 \rightarrow -(x_c - x_v) \geq r_c + r_v + 1 \\
&\qquad p_{cv} + q_{cv} >= 1
\end{aligned}
\tag{4.3}
$$

4.3 (WIP) My VCR Detection Algorithm

4.4 (WIP) C1P Detection - ILP vs PQTree Performance Comparison

Chapter 5

Voter Candidate Range Profiles

In this section we explore VCR profiles domain and describe its found properties (WIP: properties == Cechy/zalezności statystyczne). The main goal is to compare size of the domain with other well known domains such as VR and CR.

5.1 Domain analysis

We've performed several experiments. (WIP: tense: present perfect vs past simple vs present?) In each of them we check how many profiles have VCR property but not VR nor CR. For this we use the detection algorithm described in the section 4.2. For the first one we use every combination of the profiles for the given election size as the input. (WIP: czy to na pewno kombinacje i czy nie jest to zbyt pomieszane?) For the second experiment we generate random VCR profiles.

All the resources and implementation details related to the task are included in the appendix **??**.

(WIP: czy w ogóle pisać o szczegółach implementacyjnych)

(WIP: Jak często rozbijać tekst na akapity?)

5.2 (WIP) How many VCR profiles are not VR nor CR?

Experiment 1 - small profiles

For the small profiles we performed a complete search of the domain. We checked the elections with the following sizes (M candidates N voters): $[(3,3),(4,4),(5,5),(4,6),(6,4)]$. The results as seen in the table 5.1 show that VCR domain is relatively large, however only limited number of profiles are not VR nor CR.

	3C 3V	4C 4V	5C 5V	4C 6V	6C 4V
Count	2^9	2^{16}	2^{25}	2^{24}	2^{24}
VCR	506	57832	19846082	10893736	10893736
VR	-	-	-	-	-
CR	-	-	-	-	-
VCR !COP	0	96	501000	144480	144480

Table 5.1 Temporary table - Small profiles NCOP/VCR

WIP : More advanced conclusions?

WIP : Prettify the table

WIP : Make some charts, write statistics, show examples

Profile examples

Example of 5C 5V election that is VCR but not VR nor CR.

WIP: Replace with a drawing and explain why not VR/CR

```
Approval Set
    [1., 1., 0., 0., 1.]
    [1., 1., 1., 1., 1.]
    [1., 0., 0., 0., 0.]
    [0., 0., 0., 1., 1.]
    [0., 0., 1., 0., 1.]

Candidates:
    Candidate(id='A', x=3.0, r=1.0),
    Candidate(id='B', x=2.0, r=0.0),
    Candidate(id='C', x=0.0, r=0.0),
    Candidate(id='D', x=1.0, r=0.0),
    Candidate(id='E', x=1.0, r=1.0)

Voters:
    Voter(id='V0', x=2.0, r=0.0),
    Voter(id='V1', x=0.0, r=3.0),
    Voter(id='V2', x=3.0, r=0.0),
    Voter(id='V3', x=1.0, r=0.0),
    Voter(id='V4', x=0.0, r=0.0)
```

Experiment 2 — large profiles

For the larger size of the profiles we generated a suitable number of random VCR profiles and again checked how many of them are not VR nor CR. To create random VCR profile we generated points x_a and radii r_a for every agent $a \in C \cup V$ and then created preference matrix based on the VCR definition.

Additionally, to have a broader view of the domain the experiment is split into sub-experiments based on the different random distributions (WIP: Czy distributions jest ok?) used to generate profiles.

Uniform distribution

For this sub-experiment we generated points and radii with uniform distribution from the ranges $[0, 750)$ and $[0, 100)$ respectively.

(WIP: Napisać jaki wpływ mają parametry na "strukturę" profilu -> np dla 20x20. $sum(sum(A)) \in [180, 250]$)

(WIP: większy zbiór danych)

	10C 10V	15C 15V	20C 20V
VCR	1000	1000	3000
VR	-	-	-
CR	-	-	-
VCR !COP	24	667	2928

Table 5.2 Temporary table - Large profiles NCOP/VCR

Some other distribution

WIP:

Chapter 6

HD-Minimax Scoring Rule Algorithms

6.1 Hamming Distance Minimax Approval Voting

Hamming Distance

Given two sets X, Y their Hamming distance is $d_{Ham}(X,Y) = |X \setminus Y| + |Y \setminus X|$. Alternatively for a vector representation we define the Hamming distance between vectors of the same size a, b as $d_{Ham}(a,b) = |a \oplus b|$.

Minimax Approval Voting (MAV)

Scoring rule introduced by Brams et al. (2007). Selects a committees W that minimizes the maximum Hamming distance between W and voters' approval ballots, i.e. *MAV mininimizes* $max_{v \in V}\ d_{Ham}(A(v), W)$. Intuitively MAV follows the egalitarian principle (WIP: Co to dokładnie / czy istnieje definicja?), it selects committees that minimize the dissatisfaction of the least happy voter.

(WIP: poprawić zdanie)

(WIP) Computational complexity

Computing the winning committee for MAV is known to be NP-hard, see **?** (WIP: cytacje). Nevertheless, Liu and Guo (2016) provide polynomial-time algorithms for computing MAV results, for the special case of the elections with candidate range or voter range properties.

(WIP: Poprawić zdanie)

6.2 (WIP) My algorithm for VCR profiles

(WIP) Current approches

Greedy

1. Sort by ?
2. ?

Dynamic

1. Recursive
2. branch and bound
3. ?

Augumented Data Structure

1. ?

Chapter 7

(WIP) Approval deletion for VCR profile to become VR or CR

Chapter 8

Control Problem

8.1 Overview

8.2 Control for the VCR profiles

Chapter 9

Evaluation

9.1 Conclusions

9.2 Future Work

Bibliography

Booth, K. S. and Lueker, G. S. (1976). Testing for the consecutive ones property, interval graphs, and graph planarity using pq-tree algorithms. *Journal of Computer and System Sciences*, 13(3):335 – 379.

Brams, S., Kilgour, D., and Sanver, M. R. (2007). *A Minimax Procedure for Negotiating Multilateral Treaties*, pages 265–282.

Elkind, E. and Lackner, M. (2015). Structure in dichotomous preferences. *CoRR*, abs/1505.00341.

Faliszewski, P. (2021). An analysis of approval-based committee rules for 2d-euclidean elections.

Faliszewski, P., Skowron, P., Slink, A., and Talmon, N. (2017). *Multiwinner Voting: A New Challenge for Social Choice Theory*. AI Access.

Liu, H. and Guo, J. (2016). *Parameterized Complexity of Winner Determination in Minimax Committee Elections*, page 341–349. AAMAS '16. International Foundation for Autonomous Agents and Multiagent Systems.